No. of Functional Features Included in the Solution

Functional Features Overview: This document lists the functional features included in the ComicCraft project solution. The features are derived from the project description and implementation workflow, covering user input, AI story generation, image generation, preview, PDF export, API integration, and deployment.

Functional Features

Feature Summary:

Feature Category Breakdown:

Implementation Notes:

• ComicCraft uses FastAPI for backend routing and request processing.

• Gemini Flash is used for structured comic panel outlines, while Gemini Pro is used for detailed narration and dialogue.

• Stable Diffusion is used for comic-style image generation.

• Jinja2 templates provide dynamic rendering of the homepage, comic preview, and export-success pages.

• FPDF is used to compile the generated comic into a multi-page PDF.

• The current project focuses on single comic generation and does not include user account management or a database.

Team ID: SWTID-2026-3296

Date	29 September 2026
Team ID	SWTID-2026-3296
Project Name	ComicCraft - AI Comic Story Creator using Gemini Models
Maximum Marks	5 Marks

S.No	Feature Name	Feature Description	Module / Component	Status	Marks	Contribution
1	Comic Prompt & User Input	Collects story prompt, character name, setting, story tone, and preferred art style.	index.html / Frontend	Done	0.60	Enables personalized comic creation.
2	Structured Comic Outline Generation	Generates a structured 5-panel comic outline with panel number, title, scene description, and image prompt.	gemini_flash.py / Gemini Flash	Done	0.60	Creates a consistent 5-panel story structure.
3	Story Narration & Dialogue	Expands the panel outline into full comic narration and character dialogues.	gemini_pro.py / Gemini Pro	Done	0.60	Provides complete narration and dialogue.
4	AI Comic Illustration Generation	Creates comic-style illustrations from image prompts using Stable Diffusion.	image_generator.py / Stable Diffusion	Done	0.60	Adds visual illustrations for each panel.
5	Comic Panel Layout & Preview	Matches generated images with panel stories and displays the complete comic panel-by-panel.	layout_builder.py / comic_preview.html	Done	0.60	Combines visual and textual content for review.
6	PDF Comic Export	Compiles panel images, titles, descriptions, captions, and narration into a downloadable multi-page PDF.	exporters.py / FPDF	Done	0.60	Provides a shareable/offline comic output.
7	FastAPI Routes & JSON API	Provides browser and API routes including /generate, /generate-comic/json, /export-success, and /test-image.	routes.py / FastAPI	Done	0.60	Supports browser and direct API workflows.
8	Dynamic Jinja2 Frontend & Export Confirmation	Dynamically renders input, preview, and export-success pages using Jinja2 templates.	Jinja2 / HTML / CSS	Done	0.60	Provides a smooth dynamic user experience.

Metric	Count / Value
Total Features Planned	8
Total Features Implemented	8
Core / Must-Have Features	6
Additional / Nice-to-Have Features	2

Features Tested & Verified	8

S.No	Category	Features in Category	Example Features
1	User Interface (UI)	2	User input form; comic preview; export success page
2	Backend / Logic	3	Outline, story, image and layout processing
3	Database / Storage	0	No database/user-account management in current version
4	API / Integration	2	Gemini API; Stable Diffusion; FastAPI JSON endpoint
5	Security / Authentication	1	API keys handled through environment variables